

Neil deGrasse Tyson discusses Isaac Newton



<https://www.youtube.com/watch?v=danYFxGnFxQ>

Recycling of E-Waste



<https://www.youtube.com/watch?v=q78r6XCFypg>

Selection of Motor and Transmission

ME 350 Lecture

With slides from Shorya Awter

Sample Parts to Pass Around

- Gears: plastic and metal
- Hub designs for gears

Announcements

- Labs #6 and #7 (this Thursday/Friday/Monday and next Thursday/Friday) will be held in the X50 Assembly Room (across the hall from the Machine Shop) unless otherwise noted by your lab section instructor.
 - Monday Lab will not be held on October 13th due to Fall Study Break. Complete the Functional Testing on your own, outside of class.
- Lab #8 and all of the remaining lab sessions will be held in the Mechatronics Lab, room 1345 GGB.
- Reimbursement for project materials: Save all receipts. The procedure will be provided next week.

Agenda

Now that we know how a DC motor works, what its mathematical model is, and how a transmission affects that model, how do we select a DC motor and transmission for a given application?

Motor Selection

- First determine what kind of motor would be appropriate for your application, based on
 - What are the general force and power requirements?
 - What are the energy sources?
 - Cost constraints?
 - Velocity or position control required?
- Next, estimate your requirements for torque at particular speeds



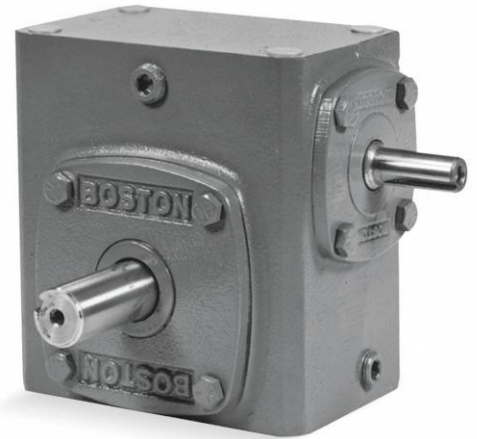
Motor Selection (continued)

- Then plot these operating conditions on a Torque-Speed curve
 - Plot as many operating points as possible
 - Make sure your units are consistent
 - Make a desired torque-speed line that would meet the above requirements
- Then plot the desired operating conditions on a Power-Speed or Power-Torque curve



Transmission Selection

- Considerations could include:
 - Gear ratio that is needed
 - Cost
 - Efficiency
 - Space requirements
 - Shock and impact loads in your application
 - Resolution (if using position control)
 - Load-Inertia Matching



Motor Selection: Example Problem

You need to select a brushed DC motor for a certain application. The motor is required to operate in the following three conditions:

- a. 2000 rpm, 70oz-inch
- b. 4000 rpm, 50oz-inch
- c. 6000 rpm, 10oz-inch

At each of these three operating conditions, it is required that there is at least a 10% torque safety factor.

1. What is your desired motor torque-speed curve to meet the above requirements? Plot the desired torque-speed curve along with the three operating points, clearly illustrating the 10% torque safety margin.

Motor Selection: Example Problem

2. Next, you have been asked to work with a given motor that has the following characteristics:

Stall Torque (oz-in)	60
No-load Speed (rpm)	16000

Will this motor do the job and meet the above design requirements stated in part 1 above? Please justify.

LECTURE EXAMPLE

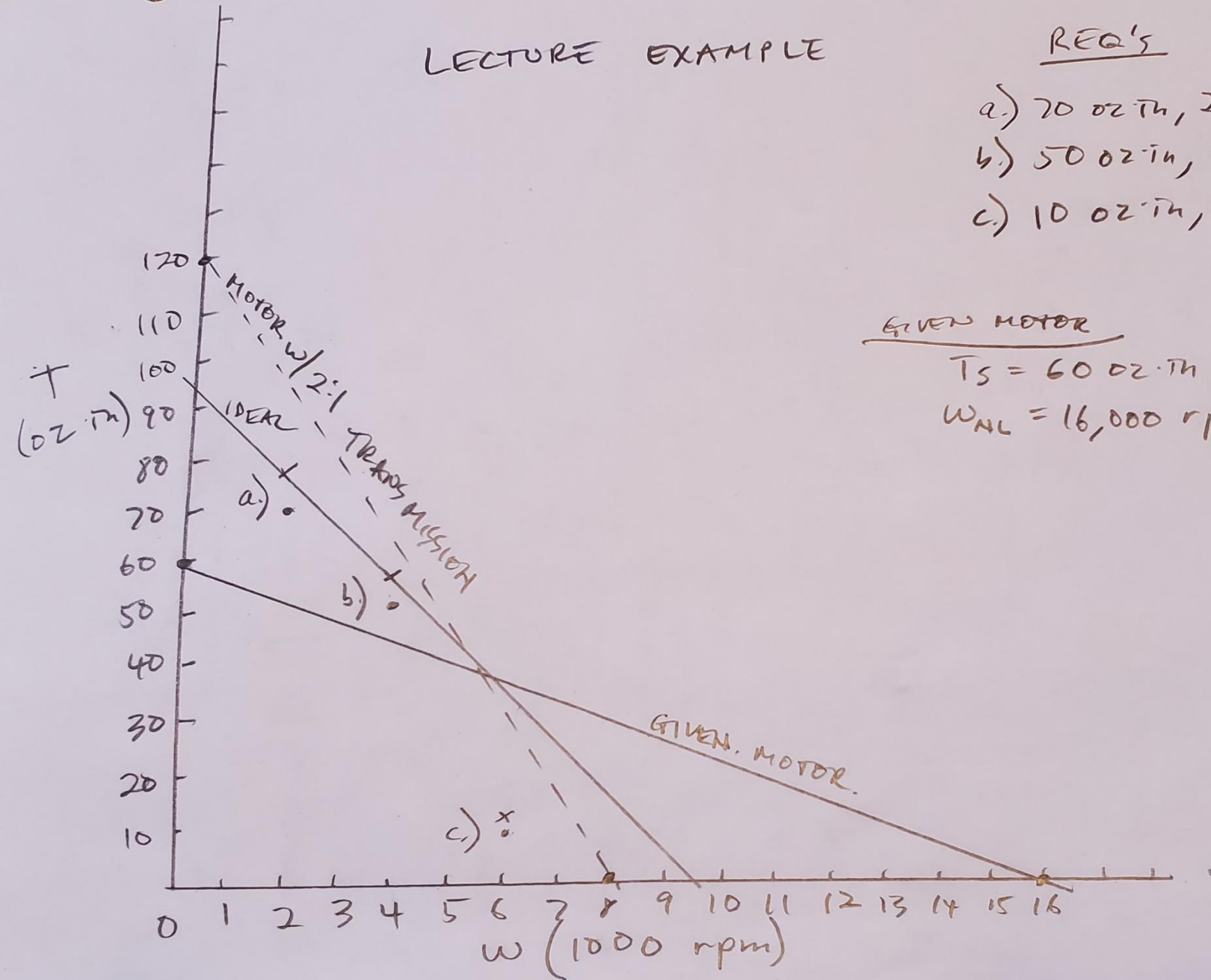
REQ'S

- a) 70 oz·in, 2000 rpm
- b) 50 oz·in, 4000 rpm
- c) 10 oz·in, 6000 rpm

GIVEN MOTOR

$$T_s = 60 \text{ oz·in}$$

$$\omega_{NL} = 16,000 \text{ rpm}$$



Motor Selection: Example Problem Solution (continued)

The given motor will not meet the requirements because operating points A and B are located above the operation line.

However, if a 2:1 gear ratio were used, it could meet the requirements.

Motor Selection: Example Problem

3. Using units of Watts, plot the required power associated with each of the three operating conditions. (When you do this, it is OK to calculate the required power for the 3 operating points without the 10% margin.)

(5)

$$P_a = (70 \text{ oz}\cdot\text{in}) (2000 \text{ rpm}) \left(\frac{0.00706 \text{ Nm}}{1 \text{ oz}\cdot\text{in}} \right) \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right) \left(\frac{1 \text{ min}}{60 \text{ sec}} \right)$$

$$= \underline{\underline{103 \text{ W @ 2000 rpm}}}$$

$$P_b = (50 \text{ oz}\cdot\text{in}) (4000 \text{ rpm}) (0.00706) \left(\frac{2\pi}{60} \right)$$

$$= \underline{\underline{148 \text{ W @ 4000 rpm}}}$$

$$P_c = \underline{\underline{44 \text{ W @ 6000 rpm}}}$$

$$P_{\text{max, GIVEN MOTOR}} = \frac{1}{4} (T_s) (\omega_{NL})$$

$$= \frac{1}{4} (60 \text{ oz}\cdot\text{in}) (16,000 \text{ rpm}) (.00706) \left(\frac{2\pi}{60} \right)$$

$$= \underline{\underline{177 \text{ W}}}$$

P
(W)

180
170
160
150
140
130
120
110
100
90
80
70
60
50
40
30
20
10

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
 ω (rpm)

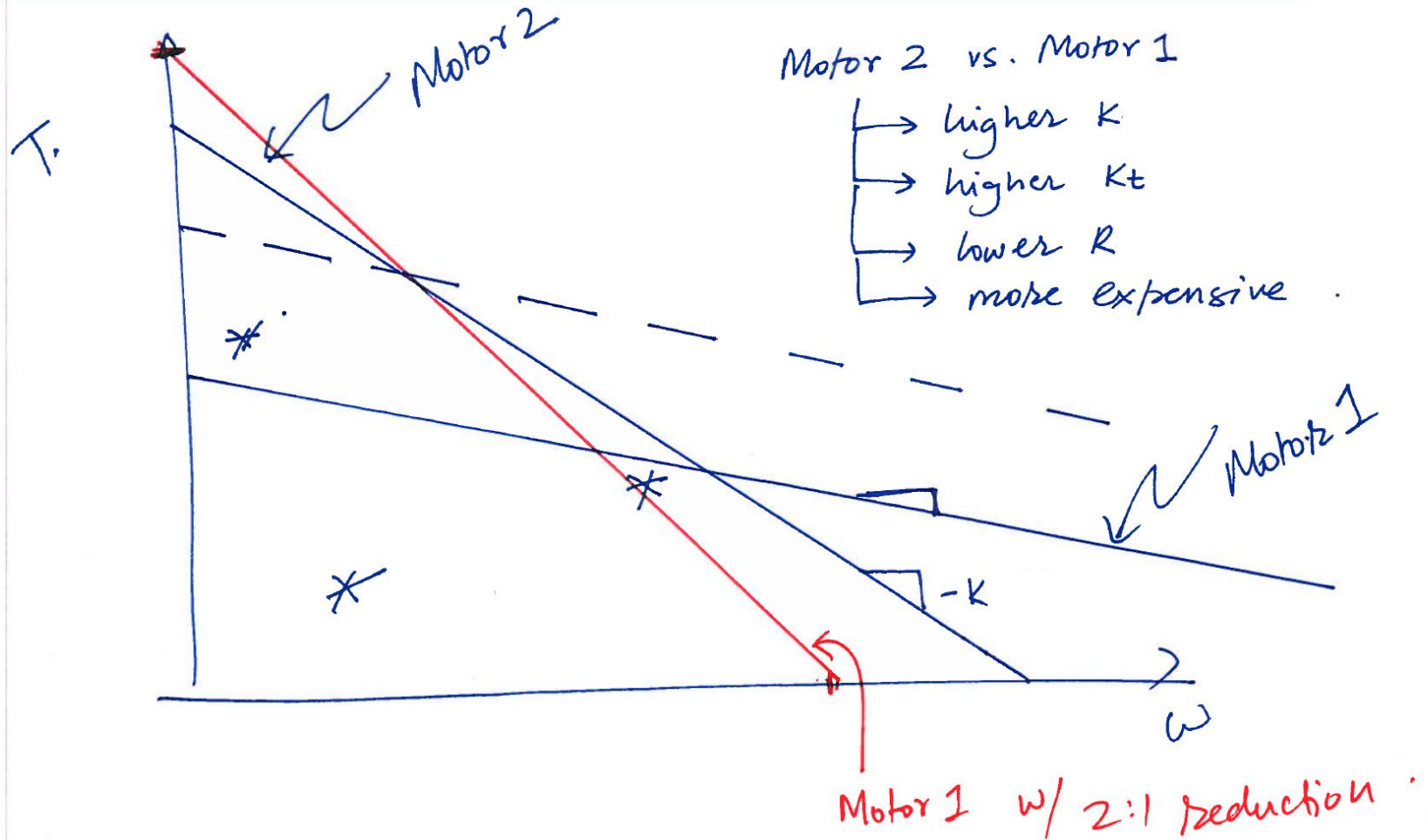
a) •

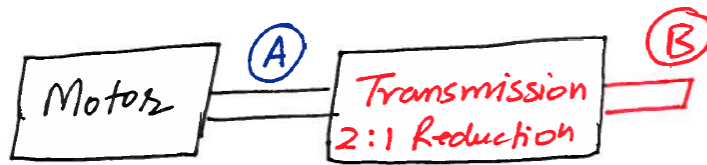
b) •

c) •

General Solution
(not for this example)

Choice of Motor with higher K versus the use of Transmission (speed reduction) to capture all desired operating points.





Transmission does not amplify power or efficiency

Typical transmission efficiency $\sim 90\%$

Impact of Transmission (2:1 reduction) on Motor Torque, Power and Efficiency, all plotted versus Motor Speed

